

# Project Proposal

CS 3743 | Database Systems

Osiris

Group 6

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## Shelter & Adoption

### Overview

For our project, we are modeling the workflow of a pet adoption shelter, from intake and medical care to applications and adoption. For why we chose this particular subject, it's simple: we like animals.

### Data Models: Entities

The schema for our database centers on animals within a pet adoption shelter and extends to people, applications, medical records, and adoption events. Each entity below includes its role in our database system and its attributes.

#### Pet

**Purpose:** Records of each pet in the shelter obtained from Kaggle with a few derived fields.

**Attributes:** PetID (PK), Name, PetType, Breed, AgeMonths, Color, Size, WeightKg, AgeMonths, Status, TimeInShelterDays, IntakeDate, PreviousOwner, AdoptionFee

#### Applicant

**Purpose:** People who apply to adopt animals (generated synthetically to provide realistic identities and contact info).

**Attributes:** ApplicantID (PK), FirstName, LastName, Email, Phone, Address

#### Application

**Purpose:** Intake and review of adoption requests for specific animals

**Attributes:** ApplicationID (PK), ApplicantID (FK), PetID (FK), SubmissionDate, Status

#### Adoption

**Purpose:** Finalized placement record for an animal with an approved applicant; enforces one adoption per animal.

**Attributes:** PetID (FK), ApplicantID (FK), ApplicationID (FK), AdoptionDate

#### Medical Record

**Purpose:** Medical history per pet (exams, vaccinations, medications, surgeries)

**Attributes:** RecordID (PK), PetID (FK), VisitDate, RecordType, Diagnosis, Treatment, VaccineType

## Data Models: Business Rules

### Business Rules

- One **Medical Record** per **Pet**
- One **Adoption** per **Pet**

### Entity-Relationships

- **Applicant** creates **Application**
- **Pet** has **Medical Record**
- **Application** creates **Adoption**
- **Adoption** has **Pet**

## Data Acquisition

For this project, we utilized [Mockaroo](#) to generate the data used within the database. Mockaroo was used to create synthetic data by defining lists of possible values for each data category. For example, a list of possible pet breeds could include German Shepherd, Poodle, and Pug, with Mockaroo randomly selecting an entry from the list for each generated record.

Synthetic data was generated for each table to provide a sufficiently varied dataset for developing and testing the database. Each table contains 100 records, allowing the database's functionality and relationships to be tested using a consistent volume of data.

The generated data was structured to maintain the necessary relationships between tables. All foreign key references correspond to existing records in their respective parent tables, ensuring that the database maintains referential integrity and does not contain invalid foreign key references.

This approach allowed the project team to create a controlled, fictional dataset that could be used to test database functionality, relationships, queries, and operations without relying on real personal or sensitive information.

# | Team Agreement Document

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## Team Name:

Osiris

## Team Members:

1. Makayla Lambert
  2. Aaron Costello
  3. Alexander Anderson
  4. Miles Martinez
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## Introduction:

The success of this group relies on the cooperation and professionalism of all members. Effective teamwork involves understanding what needs to be done, finding and assessing information, sharing the workload, and resolving any interpersonal conflicts that arise. Employers expect you to be proficient in group work, and this project will help develop those skills.

## Key Expectations for Collaboration:

- **Active Participation:** Contribute fully and genuinely to the group effort.
- **Professional Conduct:** Engage in civil discourse and adhere to academic honesty.
- **Responsibility:** Complete assigned tasks on time and to the best of your ability.
- **Accountability:** Accept the consequences of not following the group's rules.
- **Fair Recognition:** Credit group members appropriately, and do not give credit where it isn't due.

After reviewing this agreement, each member is required to sign at the end to confirm their understanding and commitment.

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# Team Agreement:

## 1. Group Goal:

**The goal of the group for this project is:**

We plan to create a database for an adoptive shelter for animals.

## 2. Meeting Schedule:

- **Meeting Frequency:** The group will meet Monday afternoons once a week online.
- **Meeting Duration:** Each meeting will last a maximum of an hour, we will be mostly working asynchronously and recombining regularly to access which areas need more attention (though this could be subject to change).

## 3. Communication:

- **Primary Communication Tool:** The main tool for group communication will be Discord.

## 4. Meeting Attendance:

- **Punctuality:** Each member agrees to arrive on time for group meetings.
- **Absences:** Members should inform the group in advance if they cannot attend a meeting due to unavoidable events like illness or other significant reasons.
- **Late Arrival:**
  - If a member is less than five minutes late, they may join quietly without disrupting the meeting. Group members are not obligated to catch up with the latecomer.
  - Consequence for Avoidable Lateness: being noted down and reported when credit is given at end of project.

## 5. Participation and Preparation:

- **Meeting Completion:** Members will remain in meetings until all tasks are completed or until a unanimous decision to adjourn is made.
- **Preparation:** Members are expected to come to meetings prepared by:
  - Reading assigned materials.
  - Bringing ideas relevant to the tasks and decisions at hand.
- **Task Completion:** Tasks must be completed by the agreed deadline. If there is an issue meeting a deadline, the member should seek help from the group in a timely manner.

## 6. Evaluation and Feedback:

- **Session Review:** At the end of each meeting, the group will evaluate its effectiveness and ensure all tasks are completed adequately.
- **Rule Enforcement:** Any member can point out if rules are being broken.

- **Academic Honesty:** If a member submits plagiarized work or cheats, the group will immediately inform the instructor.

## 7. Respect and Conduct:

- **Mutual Respect:** Treat each other with respect, avoiding name-calling and addressing ideas, not people.
- **Inappropriate Behavior:** Consequence for inappropriate behavior: being noted and reported at end of semester, if behavior is completely inexcusable they will be removed from the group.
- **Cross-Talking:** No interrupting others when they are speaking.
- **Dominating the Conversation:** The timekeeper will politely ask dominating members to allow others to speak.

## 8. Project Violations:

Violations that may lead to project failure include:

- Insufficient communication (e.g., lack of responses, missing updates).
- Delaying tasks or procrastinating on critical components.
- Failing to follow the project timeline and milestones.
- Accumulating technical debt by taking shortcuts or writing poor-quality code.
- Poor documentation of code.
- Neglecting research before implementing solutions.
- Ignoring constructive feedback.

### Consequences:

- First Violation: Inform the instructor and ask the member to rewrite any code that was not made up to standard.
  - Second Violation: Dismissal from the group.
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## 9. Roles and Responsibilities:

- **Initial Roles:** Assign roles based on individual strengths, roles may include:
    - Developer (all members should contribute).
    - Designer: Alex
    - Tester: Miles
    - Project Manager: Aaron
    - Documentation Specialist: Makayla
    - Communication Coordinator: Alex
  - **Role Flexibility:** Roles may rotate, and assignments should not impact grades.
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## 10. Project Timeline and Milestones:

- **Timeline:**
- **Sept 29** ERD complete
- **Oct 6** Dataset ready for use in SQL
- **Oct 13** Finalize division of labor for the database schema
- **Nov 17** Complete database schema
- **Nov 24** Complete demo video
- **Dec 1** Verify SQL meets all project requirements
- **Dec 5** Package everything into sql schema for submission

## 11. Additional Rules:

- [none].
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## Divorce Clause:

The group leader will evaluate each member's contributions. If a member causes ongoing dysfunction, the leader and the member(s) will meet with the instructor. If the member(s) fails to meet group expectations, they may receive a grade of zero for the project and be removed from the group. The instructor has the final say in these decisions.

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## Commitment:

We, the team named **Osiris**, agree to abide by the terms outlined in this agreement and commit to working towards the successful completion of our project.

## Signatures:

- Team Member: Makayla Lambert
- Team Member: Aaron Costello
- Team Member: Miles Martinez
- Team Member: Alexander Anderson
- Date: 9/26/2025